EARLY WARNING AND COMMUNITY-BASED EMERGENCY RESPONSE PREPAREDNESS FOR GLOF RESILIENCE IN THAME AND DOWNSTREAM VILLAGES

Installation, Verification and Community Consultation progress report

Thame Valley, Khumbu, Nepal | Spring 2026

Submitted to: *Sagarmatha Pollution Control Committee (SPCC)*
Submitted by & Date: *Tenzing Chogyal Sherpa (29 June 2026)*
Role: *Consultant Glaciologist*

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CRYOSPHERIC SOCIETY OF NEPAL

List of Abbreviations

Abbreviation	Full Form
AWS	Automatic Weather Station
RLS	Radar Level Sensor
HS	Hydrological Station
EWS	Early Warning System
masl	Metres above sea level
GLOF	Glacial Lake Outburst Flood
KPLRM	Khumbu Pasang Lhamu Rural Municipality
RTS	Real Time Solutions
AHF	American Himalayan Foundation
CSN	Cryosphere Society of Nepal
SPCC	Sagarmatha Pollution Control Committee

1. Executive Summary

This report documents the installation, verification, and community consultation activities carried out in the Thame Valley during Spring 2026. The primary goal was to deploy a complete Early Warning System (EWS) capable of alerting communities to the risk of Glacial Lake Outburst Floods (GLOF).

Ten installations were completed across the valley: two Automatic Weather Stations (AWS), three Hydrological Stations (HS), and five community warning sirens. Each was installed and verified at the locations identified in a prior site selection survey conducted in November 2025.

Following the technical installation, a community consultation and capacity-building workshop was held in Thame village. The session brought together local residents, local groups, local government and national park representatives to explain how the system works, what each component does, and how people should respond when the sirens activate. The goal was simple: to ensure that the communities most at risk from GLOFs are informed, prepared, and confident in using the system when it matters.

This work is part of the project *"Early Warning and Community-Based Emergency Response Preparedness for GLOF Resilience in Thame and Downstream Villages"*, funded by the American Himalayan Foundation (AHF) and implemented in collaboration with the Khumbu Pasang Lhamu Rural Municipality (KPLRM) and the Cryosphere Society of Nepal (CSN).

Despite administrative challenges, permit delays, and changes to the original itinerary, the field team, working closely with Real Time Solutions (RTS) and the Sagarmatha Pollution Control Committee (SPCC) successfully completed all planned installations and follow-up activities.

2. Background and Rationale

On 16 August 2024, a series of Glacial Lake Outburst Floods struck the Thame Valley and the downstream communities of Khumbu Pasang Lhamu Rural Municipality in Solukhumbu District, Nepal. The floods caused widespread destruction: approximately 25 houses and guesthouses were damaged or destroyed, along with a school, a health post, a plant nursery, and a critical downstream bridge. The speed and scale of the event left communities with very little time to react.

In the aftermath, SPCC, CSN, and local stakeholders held a series of consultations to identify the most practical and effective risk-reduction measures. Large-scale engineering works, such as lake drainage or reinforced barriers, were considered but ruled out in the short term due to the extreme remoteness, high altitude, and significant cost involved. An Early Warning System was identified as the most impactful and immediately achievable intervention. While an EWS cannot prevent a GLOF from occurring, it can provide the crucial advance notice that allows people to move to safety. It is those extra minutes that save lives.

A site selection survey was carried out in November 2025 to identify the best locations for monitoring equipment and sirens across the valley. That survey established the technical foundations for the installation work documented in this report.

3. Objectives

The key objectives of this field mission were:

- To verify and follow up with the supplier on the manufacturing and procurement of all EWS sensors and equipment
- To support the on-site installation of Automatic Weather Stations, Hydrological Stations, and community warning sirens at the locations identified in the site selection survey
- To verify that each installation was functioning correctly and transmitting data

- To hold a community consultation and capacity-building workshop in Thame to ensure local residents understand and are prepared to respond to the EWS
- To consult with local representatives and KPLRM on emergency response protocols and next steps

4. Automatic Weather Stations (AWS)

Two Automatic Weather Stations have been successfully installed at the locations identified during the November 2025 site selection survey. Each station is enclosed within protective fencing to prevent interference from people or animals, and both the AWS were confirmed to be operational and transmitting data at the time of last inspection.

Each AWS continuously measures:

- Air temperature and relative humidity
- Wind speed and direction
- Atmospheric pressure
- Rainfall and snowfall

This data feeds into the broader EWS and helps identify weather conditions that may contribute to flood risk, while also building a valuable long-term record of high-altitude climate conditions in the valley.

4.1 Thyangbo AWS

Elevation	4464 masl
Coordinates	27.824598°, 86.598778°
Location	Above Thyangbo village
Data transmission	Satellite-based real-time communication (Orbcomm)

Figure 1. The Thyangbo AWS installed above Thyangbo village, enclosed within protective fencing.

4.2 Thametheng AWS

Elevation	3816 masl
Coordinates	27.834532°, 86.648221°
Location	Between the volleyball court and the Himalayan Trust building, Thametheng
Data transmission	GSM (SIM-based) mobile network

Figure 2. Tenzing Chogyal Sherpa after completing verification of the Thametheng AWS.

The Thametheng station is well positioned not only for GLOF monitoring but also for everyday local use. Its real-time weather data will be especially valuable to trekkers and guides heading toward Renjo La Pass, offering a practical additional benefit to visitors and residents while remaining fully integrated with the primary EWS objectives.

5. Hydrological Stations (HS)

Three Hydrological Stations have been installed at the locations identified in the site selection survey. These stations use Radar Level Sensors (RLS) to monitor river water levels continuously, providing critical data for detecting unusual or rapidly rising flows that may indicate an upstream flood event. Each sensor is mounted on a stable structure, such as a large boulder or reinforced riverbank, to ensure accuracy and durability under harsh mountain conditions.

5.1 Thyangbo Hydrological Station

Elevation	4317 masl
Coordinates	27.822915°, 86.609283°
Site description	Mounted on a large, stable boulder downstream of Thyangbo village

5.2 Dudh Koshi Hydrological Station

Elevation	2856 masl
Coordinates	27.781010°, 86.722923°
Site description	Beside the bridge just before Jorsale, on the route from Namche Bazaar

5.3 Bhotekoshi Hydrological Station (Samde)

Elevation	3461 masl
Coordinates	27.823471°, 86.670445°
Site description	Downstream from Samde village

Figure 3. (Upper left) Thyangbo HS (Upper Right) Dudhkoshi/Jorsale HS (Bottom) Bhotekoshi/Thamo HS with the radar level sensors positioned over the river channel.

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6. Community Warning Sirens

6.1 Role of Sirens within the EWS

Community warning sirens are the most direct link between the technical monitoring system and the people it is designed to protect. While the AWS and hydrological stations detect and track potential flood conditions, the sirens convert those alerts into an immediate audible warning that can prompt evacuation and protective action across the valley.

Each siren has an effective audible range of approximately two kilometres under favourable conditions, making careful placement essential. The five sirens have been positioned to cover the maximum number of residents, trekkers, school children, and visitors across the valley, while also being placed away from locations where direct sound exposure could cause hearing damage.

Beyond emergency flood warnings, the sirens have broader value. When not in use for flood events, which will be the majority of the time, they can be activated by KPLRM for wider community communication, public announcements, and other municipality-led messaging. This flexibility is intended to keep the system familiar and trusted within the community, so it is not an unfamiliar device encountered only in a crisis.

Responsibility for triggering each siren has been handed over to the respective ward representatives of KPLRM. Detailed emergency response protocols will be developed and finalised in further consultation with local authorities and community members.

6.2 Siren Locations

Thame Village Siren

Elevation	3876 masl
Coordinates	27.830702°, 86.645591°
Site description	Positioned just above Thame village on the moraine

Figure 4. The Thame village siren, overlooking the valley below.

Thamo Siren:

Elevation	3,555 m above sea level
Coordinates	27.82369°N, 86.67636°E
Site description	Above Thamo village, toward the trekking route

Figure 5. The Thamo siren installed among the forest above the village.

Monjo Siren

Elevation	2843 masl
Coordinates	27.773150°, 86.721412°
Site description	Near Monjo Monastery

Figure 6. The Monjo siren, positioned to cover the village and the surrounding trekking route.

Phakding Siren

Elevation	2,665 masl
Coordinates	27.73928°N, 86.71314°E
Site description	Positioned beside the existing Imja Siren

Surke Siren

Elevation	2,268 masl
Coordinates	27.67302°N, 86.71676°E
Site description	Just above Surke village, toward the main trekking route

7. Community Consultation and Capacity-Building Workshop

A community consultation and capacity-building workshop was held in Thame village following the completion of the station installations. The event was designed to ensure that the people most directly affected by a potential GLOF, including residents, lodge owners, school staff, and local leaders, have a clear and practical understanding of the EWS, how it works, and what to do when it activates.

7.1 Objectives of the Workshop

The workshop had four main aims:

- To introduce and explain the components of the EWS in clear, accessible terms for a non-technical audience
- To walk participants through how the system detects flood risk and triggers community warnings
- To discuss and agree on emergency response protocols, including what the sirens mean, where to evacuate, and who is responsible for what
- To create an open space for community members to ask questions, raise concerns, and share their own knowledge of flood risk in the valley

7.2 Participation and Engagement

The workshop was well attended, with community members from Thame and surrounding areas joining KPLRM ward representatives and the project team. The turnout reflected a genuine appetite within the community for understanding the system that is now in place to protect them.

Discussion was open, substantive, and productive throughout the session. Participants asked detailed and thoughtful questions about how the sensors work, how quickly the sirens would activate after a flood is detected, and what the expected lead time for evacuation would be. Community members also shared their own firsthand experiences of past flood events, which added valuable local knowledge to the conversation and helped shape the discussion on response protocols.

There was broad support for the EWS among those present. Participants expressed a strong sense of ownership over the system and a clear desire to see it maintained and used effectively for the long term. Several attendees requested follow-up training sessions, and the project team committed to providing ongoing support.

Figure 7. Workshop proceedings in Thame: (Top) A National Park representative addressing the participants; (Middle) Tenzing Chogyal Sherpa presenting on the 2024 Thame GLOF and the Early Warning System; (Bottom left) Pasang Ngima Sherpa of SPCC presenting on the EWS and facilitating the session (Bottom right) Representative from Thame village presenting his views.

Figure 8. (Top left) Pasang Gyalzen Sherpa, ward leader for Ward 5, KPLRM presenting opening remarks (Middle left) Participating interacting and discussing during the demonstration of the Siren at Thame; (Middle Right) Pasang Ngima Sherpa of SPCC handing over the remote to ward leader – Pasang Ngima Sherpa. (Bottom) Overview of the Siren demonstration.

7.3 Key Topics Covered

- Overview of GLOF risk in the Thame Valley, with reference to the August 2024 flood event and its impacts
- How the Automatic Weather Stations and Hydrological Stations collect and transmit data in real time
- How data thresholds trigger alerts and activate the sirens
- Siren tone meanings and the expected community response, including evacuation routes and safe assembly points
- Roles and responsibilities of KPLRM ward members as designated siren operators
- The importance of regular system checks and community practice drills
- How the sirens can also be used for broader public communication by local authorities
- **Hands on demonstration** and training of the Siren installed above Thame.

7.4 Outcomes

By the end of the workshop, participants had a comprehensive overview of the EWS and felt confident about their role within the region. The session produced clear and actionable outcomes. Ward representatives confirmed their understanding of the responsibility to activate sirens when required and the local community members have also had positive responses towards it.

The workshop reinforced a fundamental principle: the technical components of an EWS are only part of what makes a warning system effective. Community trust, preparedness, and clear communication are equally important, and this session took meaningful steps toward building all three.

8. Summary of Installed Stations

The table below provides a complete overview of all EWS components installed and verified during the Spring 2026 field mission.

Table 1. All EWS stations installed in the Thame Valley, Spring 2026.

Station Name	Latitude	Longitude	Elevation (masl)	Type
Thyangbo AWS	27.824598°	86.598778°	4464	AWS
Thametheng AWS	27.834532°	86.648221°	3816	AWS
Thyangbo RLS	27.822915°	86.609283°	4317	HS
Dudh Koshi RLS (Jorsale)	27.781010°	86.722923°	2856	HS
Bhotekoshi RLS (near Samde)	27.823471°	86.670445°	3461	HS
Thame Siren	27.830702°	86.645591°	3876	Siren
Monjo Siren	27.773150°	86.721412°	2843	Siren
Phakding Siren	27.740964°	86.713379°	2665	Siren
Surke Siren	27.672789°	86.716268°	2262	Siren
Thamo Siren	27.823693°	86.676360°	3555	Siren

9. Conclusions and Next Steps

The Spring 2026 field mission marks a significant milestone for the "*Early Warning and Community-Based Emergency Response Preparedness for GLOF Resilience in Thame and Downstream Village*" project. All ten components of the Early Warning System, comprising two Automatic Weather Stations, three Hydrological Stations, and five community sirens, have been successfully installed, verified, and confirmed as operational. The follow-up community workshop in Thame ensured that this technical achievement translates into genuine preparedness on the ground.

For the first time, the Thame Valley now has the capacity to detect a rapidly developing flood situation and alert communities in time to take protective action. This is a meaningful step forward for the safety and resilience of the Khumbu region, and a direct response to the devastating events of August 2024.

Several important workstreams remain, and the project team is committed to seeing them through:

- Finalising the emergency response protocol with KPLRM, translating agreed procedures into a clear, printed document that ward representatives and community members can keep and refer to
- Completing a formal handover and sustainability agreement with KPLRM that defines ownership, maintenance responsibilities, and long-term management arrangements for each station
- Conducting further capacity-building training for SPCC staff and other relevant personnel, covering system maintenance, data interpretation, and basic troubleshooting
- Producing and distributing community information materials, including simple response guides and evacuation maps, to households, schools, and lodges throughout the valley

What has been built in Thame is not simply a set of sensors and sirens. It is a system that, when combined with informed and prepared communities, has the potential to save lives. The work ahead will focus on embedding that system into everyday community life, ensuring it remains well maintained, well understood, and ready to perform when it is needed most.

Report prepared by Tenzing Chogyal Sherpa, Consultant Glaciologist | Spring 2026